

1 Parametrized Curves

1.1 Parametrized Curves

Definition 1. (Parametrized Curve)

A *parametrized curve* in $\mathbb{R}^n$ is a map

$$\gamma : (\alpha, \beta) \longrightarrow \mathbb{R}^n,$$

where $\alpha, \beta \in \mathbb{R} \cup \{\pm\infty\}$.

Example 1.(Parabola) We can give the parabola $y = x^2$ a parametrization by $\gamma(t) := (t, t^2)$ or $\gamma(t) := (2t, 4t^2)$ or $\gamma(t) = (t^3, t^6)$ with $t \in \mathbb{R}$ and so on. So the parametrization of a given curve is not unique.

Example 2.(Circle) The circle $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = r^2\}$ can be parametrized by

$$\gamma : \mathbb{R} \longrightarrow \mathbb{R}^2$$

$$t \mapsto (r \cos t, r \sin t).$$

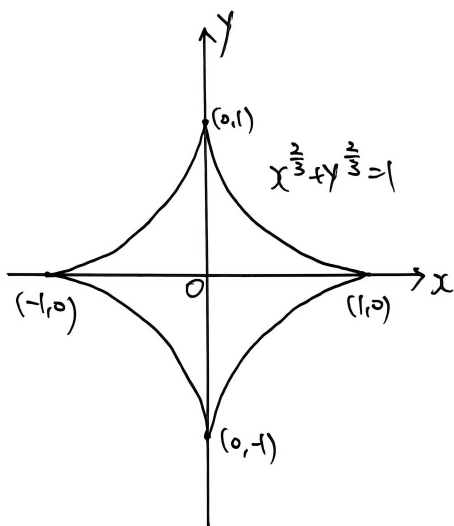
And we can note that $\gamma(t + 2\pi) = \gamma(t)$ for $\forall t \in \mathbb{R}$.

Example 3.(Astroid) The curve

$$\gamma(t) := (\cos^3 t, \sin^3 t), \quad t \in \mathbb{R}$$

is a parametrization of a curve called astroid, which implies that the coordinates $x = \cos^3 t$, $y = \sin^3 t$ satisfies the equation

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1.$$



Definition 2. For a vector-valued function

$$\gamma : \mathbb{R} \longrightarrow \mathbb{R}^n$$

$$t \mapsto (\gamma_1(t), \gamma_2(t), \dots, \gamma_n(t)),$$

we do the derivative and integration by

$$\frac{d\gamma}{dt} := \left(\frac{d\gamma_1}{dt}, \frac{d\gamma_2}{dt}, \dots, \frac{d\gamma_n}{dt} \right), \quad \int \gamma(t) dt := \left(\int \gamma_1(t) dt, \int \gamma_2(t) dt, \dots, \int \gamma_n(t) dt \right).$$

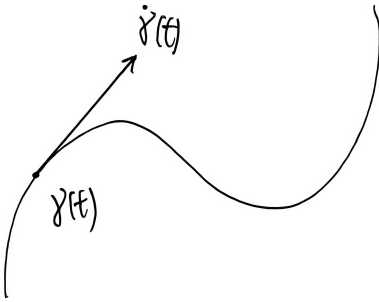
Notation: We will often denote $\frac{d\gamma}{dt}, \frac{d^2\gamma}{dt^2}$ by $\dot{\gamma}(t), \ddot{\gamma}(t)$.

Convention: We will always assume that the parametrized curve $\gamma(t)$ is smooth, i.e., $\frac{d^n\gamma}{dt^n}$ exists for $\forall n \in \mathbb{Z}^+$, if there is no specific assumption.

1.2 Tangent Vector of a Parametrized Curve

Definition 3.(Tangent Vector)

If $\gamma(t)$ is a parametrized curve, its first derivative $\dot{\gamma}(t)$ is called its *tangent vector*.



Proposition 1. If $\dot{\gamma}(t)$ is a constant vector, then the image of the parametrized curve $\gamma(t)$ is a straight line.

Proof: Suppose $\dot{\gamma}(t) \equiv \mathbf{a}$ for $\forall t$, where $\mathbf{a}$ is a constant vector, if we denote $\mathbf{a} \triangleq (a_1, a_2, \dots, a_n)$ with $a_i \in \mathbb{R}$, then

$$\begin{aligned} \gamma(t) &= \int \dot{\gamma}(t) dt = \int \mathbf{a} dt = \left(\int a_1 dt, \int a_2 dt, \dots, \int a_n dt \right) \\ &= (a_1 t + b_1, a_2 t + b_2, \dots, a_n t + b_n) \\ &= t\mathbf{a} + \mathbf{b}, \end{aligned}$$

where $\mathbf{b} \triangleq (b_1, b_2, \dots, b_n)$ is another constant vector. This is a parametric equation of a straight line whose direction parallels to $\mathbf{a}$ and passing the point $\mathbf{b}$. ■

1.3 Arc-length of a Parametrized Curve

Definition 4.(Arc-Length) The *arc-length* of a curve $\gamma(t)$ whose starting point is $\gamma(t_0)$ is the scalar function

$$s(t) := \int_{t_0}^t \|\dot{\gamma}(u)\| du.$$

Remark: So by this definition $s(t_0) = 0$ and $s(t) > 0$ or $s(t) < 0$ is depending on $t > t_0$ or $t < t_0$.

Example 4.(Logarithmic Spiral)

The logarithmic spiral is

$$\gamma(t) := (e^{kt} \cos t, e^{kt} \sin t),$$

where $k \in \mathbb{R}^*$ is a constant. Thus we get

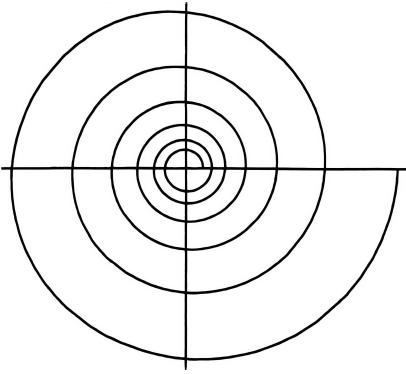
$$\dot{\gamma}(t) = (e^{kt}(k \cos t - \sin t), e^{kt}(k \sin t + \cos t))$$

therefore

$$\|\dot{\gamma}(t)\|^2 = (k^2 + 1)e^{2kt}.$$

So if we choose the starting point as $\gamma(0) = (1, 0)$, the the arc-length is

$$s(t) = \int_0^t \|\dot{\gamma}(u)\| du = \int_0^t \sqrt{k^2 + 1} e^{ku} du = \frac{\sqrt{k^2 + 1}}{k} (e^{kt} - 1).$$



Proposition 2. Let $\gamma(t)$ be a parametrized curve

(1) if $s(t)$ is the arc-length function of $\gamma(t)$ starting at $\gamma(t_0)$ and $\tilde{s}(t)$ is the arc-length function of $\gamma(t)$ starting at another point $\gamma(\tilde{t}_0)$, then $s(t) - \tilde{s}(t) = \text{constant}$;

(2) we have

$$\frac{ds}{dt} = \|\dot{\gamma}(t)\|.$$

Proof: (1) Note that

$$\tilde{s}(t) = \int_{\tilde{t}_0}^t \|\dot{\gamma}(u)\| du = \left(\int_{\tilde{t}_0}^{t_0} + \int_{t_0}^t \right) \|\dot{\gamma}(u)\| du = \int_{\tilde{t}_0}^{t_0} \|\dot{\gamma}(u)\| du + s(t),$$

which means $\tilde{s}(t) - s(t) = \text{constant}$;

(2) we have

$$\frac{ds}{dt} = \frac{d}{dt} \left(\int_{t_0}^t \|\dot{\gamma}(u)\| du \right) = \|\dot{\gamma}(t)\|,$$

which completes the proof. ■

Definition 5. The *speed* of $\gamma : (\alpha, \beta) \longrightarrow \mathbb{R}^n$ at the point $\gamma(t)$ is defined as $\|\dot{\gamma}(t)\|$. γ is called *unit-speed* if $\|\dot{\gamma}(t)\| \equiv 1$ for $\forall t \in (\alpha, \beta)$.

We have the following useful and important results in computation.

Proposition 3. Let $\mathbf{a}(t) \triangleq (a_1(t), \dots, a_n(t))$ and $\mathbf{b}(t) \triangleq (b_1(t), \dots, b_n(t))$ be two vectors in $\mathbb{R}^n$ depending on t . Then we have

$$\frac{d}{dt} (\mathbf{a}(t) \cdot \mathbf{b}(t)) = \frac{d\mathbf{a}(t)}{dt} \cdot \mathbf{b}(t) + \mathbf{a}(t) \cdot \frac{d\mathbf{b}(t)}{dt}.$$

If $\mathbf{a}(t) = (a_1(t), a_2(t), a_3(t))$ and $\mathbf{b}(t) = (b_1(t), b_2(t), b_3(t))$ are vectors in $\mathbb{R}^3$, we have

$$\frac{d}{dt} (\mathbf{a}(t) \times \mathbf{b}(t)) = \frac{d\mathbf{a}(t)}{dt} \times \mathbf{b}(t) + \mathbf{a}(t) \times \frac{d\mathbf{b}(t)}{dt}.$$

Proof: (1) Since $\mathbf{a}(t) \cdot \mathbf{b}(t) = \sum_{j=1}^n a_j(t)b_j(t)$, so we have

$$\begin{aligned} \frac{d}{dt} (\mathbf{a}(t) \cdot \mathbf{b}(t)) &= \frac{d}{dt} \left(\sum_{j=1}^n a_j(t)b_j(t) \right) = \sum_{j=1}^n \frac{d}{dt} (a_j(t)b_j(t)) \\ &= \sum_{j=1}^n \left(\frac{da_j(t)}{dt} b_j(t) + a_j(t) \frac{db_j(t)}{dt} \right) \\ &= \sum_{j=1}^n \frac{da_j(t)}{dt} b_j(t) + \sum_{j=1}^n a_j(t) \frac{db_j(t)}{dt} \\ &= \frac{d\mathbf{a}(t)}{dt} \cdot \mathbf{b}(t) + \mathbf{a}(t) \cdot \frac{d\mathbf{b}(t)}{dt}; \end{aligned}$$

(2) Since we see that

$$\mathbf{a}(t) \times \mathbf{b}(t) = \left(\det \begin{pmatrix} a_2(t) & a_3(t) \\ b_2(t) & b_3(t) \end{pmatrix}, -\det \begin{pmatrix} a_1(t) & a_3(t) \\ b_1(t) & b_3(t) \end{pmatrix}, \det \begin{pmatrix} a_1(t) & a_2(t) \\ b_1(t) & b_2(t) \end{pmatrix} \right),$$

note that from the theory of calculus and linear algebra, we know that

$$\frac{d}{dt} \left[\det \begin{pmatrix} a_{11}(t) & a_{12}(t) & \dots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \dots & a_{2n}(t) \\ \vdots & \vdots & & \vdots \\ a_{n1}(t) & a_{n2}(t) & \dots & a_{nn}(t) \end{pmatrix} \right] = \sum_{k=1}^n \det \begin{pmatrix} a_{11}(t) & a_{12}(t) & \dots & a_{1n}(t) \\ \vdots & \vdots & & \vdots \\ \frac{da_{k1}(t)}{dt} & \frac{da_{k2}(t)}{dt} & \dots & \frac{da_{kn}(t)}{dt} \\ \vdots & \vdots & & \vdots \\ a_{n1}(t) & a_{n2}(t) & \dots & a_{nn}(t) \end{pmatrix}$$

taking this into the equation that we want to compute, we can easily get the result. ■

Proposition 4. Let $\mathbf{n}(t)$ be a vector such that $\|\mathbf{n}(t)\| \equiv \text{constant}$, then we have

$$\dot{\mathbf{n}}(t) \cdot \mathbf{n}(t) \equiv 0 \quad \text{for } \forall t,$$

which means that $\dot{\mathbf{n}}(t) \perp \mathbf{n}(t)$ for all t .

Proof: Since $\|\mathbf{n}(t)\| \equiv \text{constant}$, so we get

$$\frac{d}{dt}(\|\mathbf{n}(t)\|^2) = \frac{d}{dt}(\mathbf{n}(t) \cdot \mathbf{n}(t)) = 2\dot{\mathbf{n}}(t) \cdot \mathbf{n}(t) \equiv 0,$$

which means $\dot{\mathbf{n}}(t) \cdot \mathbf{n}(t) \equiv 0$ for $\forall t$. ■

So we immediately get the following corollary.

Corollary 1. If γ is a unit-speed curve, i.e., $\|\dot{\gamma}(t)\| \equiv 1$ for $\forall t$, then $\dot{\gamma}(t) \perp \ddot{\gamma}(t)$ for all t .

Proof: By Proposition 4. ■

1.4 Reparametrization